

Tutorial - 2 - Answers

1. $\frac{1}{3}te^{-t} - \frac{1}{9}e^{-t} + \frac{1}{900}e^{-4t} + \frac{11}{100}e^{-t}(\cos t - 3\sin t) + \frac{1}{20}e^{-t}(3t\cos t - 3\sin t + t\sin t).$
2. $\frac{d^4c(t)}{dt^4} + 9\frac{d^3c(t)}{dt^3} + 126\frac{d^2c(t)}{dt^2} + 724\frac{dc(t)}{dt} + 1200c(t) = \frac{d^3r(t)}{dt^3} + 5\frac{d^2r(t)}{dt^2} + 13\frac{dr(t)}{dt} + 14r(t).$
3. $G(s) = \frac{s^2 + s + 1}{s^3 + 3s^2 + 4s + 2}.$ The steady-state value $(y_{ss}) = \frac{1}{2}.$
4. (a) $G(s) = \frac{s + 3}{(s + 2)(s + 5)}.$
 (b) $y(t) = \left(\frac{1}{2}e^{-t} - \frac{1}{3}e^{-2t} - \frac{1}{6}e^{-5t}\right)u(t).$
 (c) $y(t) = \left(\frac{1}{2}e^{-t} + \frac{3}{2}e^{-2t} - e^{-5t}\right)u(t).$
5. $\frac{I(s)}{V(s)} = \frac{96s^3 + 120s^2 + 36s + 12}{288s^4 + 360s^3 + 204s^2 + 192s + 12}.$
6. $\frac{V_0(s)}{V_i(s)} = \frac{R_1R_2C_1C_2s^2 + R_2C_1s + R_1C_1s + R_2C_2s + 1}{R_1R_2C_1C_2s^2 + R_1C_1s + R_2C_2s + 1}.$